Analytic continuation of the Riemann Zeta function

Integral Master

February 2026

1 Gathering the initial necessities

In this document, we'll need:

-The Riemann Zeta function:

$$\zeta(s) = \sum_{n=1}^{+\infty} n^{-s}$$

-The Gamma function:

$$\Gamma(z) = \int_0^{+\infty} t^{z-1} e^{-t} dt$$

-The Mellin transform:

$$\mathcal{M}\{f(x)\}(s) = \int_0^{+\infty} x^{s-1} f(x) dx$$

-The Fourier transform:

$$\hat{f}(\xi) = \mathcal{F}\{f(x)\}(\xi) = \int_{-\infty}^{+\infty} f(x) e^{-2\pi i \xi x} dx$$

-The Poisson summation formula:

$$\sum_{n=-\infty}^{+\infty} f(n) = \sum_{\omega=-\infty}^{+\infty} \hat{f}(\omega)$$

-The Jacobi Theta function:

$$\theta(t) = \sum_{n=-\infty}^{+\infty} e^{-\pi n^2 t}$$

2 Some points we need to make before the actual start of the work

First, we need to clarify the form of $\theta(t)-1$ as a sum, so:

$$\theta(t)=\sum_{n=-\infty}^{+\infty}e^{-\pi n^2 t}=(\cdots+e^{-\pi(-2)^2 t}+e^{-\pi(-1)^2 t}+1+e^{-\pi(1)^2 t}+e^{-\pi(2)^2 t}+\dots)=2\sum_{n=1}^{+\infty}e^{-\pi n^2 t}+1$$

$$\implies \theta(t)-1=2\sum_{n=1}^{+\infty}e^{-\pi n^2 t}$$

Second, we need to reach a certain relation with the theta function using the Poisson summation formula, so take $f(x)=e^{-\pi x^2 t}$, then:

$$\hat{f}(\omega)=\int_{-\infty}^{+\infty}f(x)e^{-2\pi i\omega x}dx=\int_{-\infty}^{+\infty}e^{-\pi x^2 t}e^{-2\pi i\omega x}dx=\int_{-\infty}^{+\infty}e^{-\pi x^2 t-2\pi i\omega x}dx$$

$$=\int_{-\infty}^{+\infty}e^{-\pi(x^2 t+2i\omega x)}dx=\int_{-\infty}^{+\infty}e^{-\pi t\left(x^2+2\left(\frac{i\omega}{t}\right)+\left(\frac{i\omega}{t}\right)^2-\left(\frac{i\omega}{t}\right)^2\right)}dx$$

$$=\int_{-\infty}^{+\infty}e^{-\pi t\left(\left(x+\frac{i\omega}{t}\right)^2-\left(\frac{i\omega}{t}\right)^2\right)}dx=\int_{-\infty}^{+\infty}e^{-\pi t\left(x+\frac{i\omega}{t}\right)^2}e^{-\frac{\pi\omega^2}{t}}dx$$

Let $u=x+\frac{i\omega}{t}$

$$\implies \hat{f}(\omega)=e^{-\frac{\pi\omega^2}{t}}\int_{-\infty}^{+\infty}e^{-\pi t u^2}du=e^{-\frac{\pi\omega^2}{t}}\sqrt{\frac{\pi}{\pi t}}=\frac{e^{-\frac{\pi\omega^2}{t}}}{\sqrt{t}}$$

$$\implies \theta(t)=\sum_{\omega=-\infty}^{+\infty}\frac{e^{-\frac{\pi\omega^2}{t}}}{\sqrt{t}}=\frac{1}{\sqrt{t}}\sum_{\omega=-\infty}^{+\infty}e^{-\pi\omega^2\left(\frac{1}{t}\right)}=\frac{\theta\left(\frac{1}{t}\right)}{\sqrt{t}}$$

3 The functional equation

We have

$$\mathcal{M}\{\theta(t)-1\}(\alpha+1)=\int_0^{+\infty}t^\alpha(\theta(t)-1)dt$$

$$=\int_0^{+\infty}t^\alpha\cdot 2\sum_{n=1}^{+\infty}e^{-\pi n^2 t}dt=2\sum_{n=1}^{+\infty}\int_0^{+\infty}t^\alpha e^{-\pi n^2 t}dt$$

Let:

$$u=\pi n^2 t$$

$$\implies \mathcal{M}\{\theta(t)-1\}(\alpha+1)=2\sum_{n=1}^{+\infty}\int_0^{+\infty}\left(\frac{u}{\pi n^2}\right)^\alpha\cdot e^{-u}\cdot\left(\frac{1}{\pi n^2}\right)du$$

$$= 2\sum_{n=1}^{+\infty} \pi^{-(\alpha+1)} n^{-2(\alpha+1)} \int_0^{+\infty} u^{\alpha} e^{-u} du = 2\pi^{-(\alpha+1)} \Gamma(\alpha+1)\zeta(2(\alpha+1))$$

Let

$$s = 2(\alpha+1) \implies \alpha + 1 = \frac{s}{2} \implies \alpha = \frac{s}{2} - 1$$

$$\implies \implies \mathcal{M}\{\theta(t) - 1\}\left(\frac{s}{2}\right) = 2\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right)\zeta(s)$$

$$\implies 2\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right)\zeta(s) = \int_0^{+\infty} t^{\left(\frac{s}{2}-1\right)}(\theta(t) - 1)dt$$

4 Manipulation of the integral

We have

$$\int_0^{+\infty} t^{\left(\frac{s}{2}-1\right)}(\theta(t) - 1)dt = \int_0^{1} t^{\left(\frac{s}{2}-1\right)}(\theta(t) - 1)dt + \int_1^{+\infty} t^{\left(\frac{s}{2}-1\right)}(\theta(t) - 1)dt$$

$$\int_0^{1} t^{\left(\frac{s}{2}-1\right)}(\theta(t) - 1)dt = \int_0^{1} t^{\left(\frac{s}{2}-1\right)} \left(\frac{1}{\sqrt{t}}\theta\left(\frac{1}{t}\right) - 1\right) dt$$

Let:

$$u = \frac{1}{t} \implies dt = -\frac{1}{u^2}du$$

$$\implies \int_0^{1} t^{\frac{s}{2}-1} \left(\frac{1}{\sqrt{t}}\theta\left(\frac{1}{t}\right) - 1\right) dt = \int_1^{+\infty} u^{-\left(\frac{s}{2}-1\right)} \left(\sqrt{u} \cdot \theta(u) - 1\right) \cdot u^{-2} du$$

$$= \int_1^{+\infty} u^{-\frac{s}{2}+1}((u^{\frac{1}{2}} \cdot \theta(u) - u^{\frac{1}{2}}) + (u^{\frac{1}{2}} - 1))u^{-2} du = \int_1^{+\infty} u^{-\frac{s}{2}-\frac{1}{2}}(\theta(u) - 1)du + \int_1^{+\infty} u^{-\frac{s}{2}-\frac{1}{2}} - u^{-\frac{s}{2}-1} du$$

Let:

$$I = \int_1^{+\infty} u^{-\frac{s}{2}-\frac{1}{2}} - u^{-\frac{s}{2}-1} du = \lim_{n\to+\infty} \left[\frac{u^{-\frac{s}{2}+\frac{1}{2}}}{-\frac{s}{2}+\frac{1}{2}} + \frac{u^{-\frac{s}{2}}}{\frac{s}{2}}\right]_1^n$$

$$= \lim_{n\to+\infty} \left[\frac{n^{-\frac{s}{2}+\frac{1}{2}}}{-\frac{s}{2}+\frac{1}{2}} - \frac{1}{-\frac{s}{2}+\frac{1}{2}} + \frac{n^{-\frac{s}{2}}}{\frac{s}{2}} - \frac{1}{\frac{s}{2}}\right] = \frac{2}{s-1} - \frac{2}{s}$$

$$\implies 2\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right)\zeta(s) = \frac{2}{s-1} - \frac{2}{s} + \int_1^{+\infty} t^{-\frac{s}{2}-\frac{1}{2}}(\theta(t) - 1)dt + \int_1^{+\infty} t^{\left(\frac{s}{2}-1\right)}(\theta(t) - 1)dt$$

$$= \frac{2}{s-1} - \frac{2}{s} + \int_1^{+\infty} (t^{-\frac{s}{2}-\frac{1}{2}} + t^{\frac{s}{2}-1})(\theta(t) - 1)dt$$

$$\implies \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right)\zeta(s) = \frac{1}{s-1} - \frac{1}{s} + \int_1^{+\infty} (t^{-\frac{s}{2}-\frac{1}{2}} + t^{\frac{s}{2}-1})(\theta(t) - 1)dt.$$

$$\implies \zeta(s) = \frac{\pi^{\frac{s}{2}}}{\Gamma\left(\frac{s}{2}\right)} \left[\frac{1}{s-1} - \frac{1}{s} + \int_1^{+\infty} \left(t^{-\frac{s}{2}-\frac{1}{2}} + t^{\frac{s}{2}-1}\right)(\theta(t) - 1)\, dt\right]$$

Since $\frac{1}{\Gamma(s)}$ is entire and $\theta(t)$ converges $\forall t \in [1, +\infty)$, and the integral converges $\forall s \in \mathbb{C}$ and $\frac{1}{\Gamma\left(\frac{s}{2}\right)}$ cancels the pole at 0 (because $|s| \to 0 : O\left(\frac{1}{\Gamma(s)}\right) = \frac{1}{s}$), we have isolated the single pole of $\zeta(s)$ at 1.